Hybrid RAG and DeBERTa-Based System foe Hallunications Dection in Large Model Outputes

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Abstract— Hallucination in Large Language Models (LLMs) — the generation of factually incorrect yet plausible-sounding text — poses a critical challenge for the deployment of AI systems in reliability-sensitive domains. This paper presents a comprehensive study of two hallucination detection architectures: a fine-tuned BERT-based binary classifier and a hybrid Retrieval-Augmented Generation (RAG) system incorporating DeBERTa-v3-base with FAISS-indexed knowledge retrieval and Natural Language Inference (NLI)-based explainability. The BERT baseline, trained on approximately 140,000 balanced samples using the Hugging Face Trainer API with AdamW optimization and cross-entropy loss, achieves a precision of 93.46%, recall of 94.26%, and F1-score of 93.86% under CPU-based training. The hybrid model retrieves the top-3 semantically similar knowledge base documents per claim using all-MiniLM-L6-v2 sentence embeddings and a FAISS IndexFlatIP vector store, subsequently classifying the fused (claim, context) pair with DeBERTa-v3-base. An NLI cross-encoder provides interpretable entailment verdicts for each prediction. While the hybrid model's accuracy is constrained to approximately 50% due to limited knowledge base coverage during evaluation, the architecture demonstrates the viability of explainable, retrieval-grounded hallucination detection. Experimental findings highlight the critical dependency of retrieval-based systems on knowledge base completeness and motivate future extensions including Wikipedia-scale indexing and GPU-accelerated training.

Keywords: *hallucination detection, large language models, retrieval-augmented generation, DeBERTa, BERT, FAISS, natural language inference, factual verification*

I. INTRODUCTION

Large Language Models (LLMs) such as GPT-4, LLaMA-2, and PaLM-2 have demonstrated significant advancements in natural language understanding and generation. These models are widely applied in tasks including question answering, text summarization, code generation, and decision support systems. Despite their capabilities, LLMs are prone to a critical limitation known as hallucination, where the model generates statements that are factually incorrect, unverifiable, or logically inconsistent. Such outputs are often linguistically fluent and contextually plausible, making them difficult to detect through superficial evaluation.

The impact of hallucination is particularly concerning in high-stakes domains. In healthcare, incorrect information may lead to unsafe recommendations; in legal applications, fabricated references can compromise reliability; and in financial systems, inaccurate data may result in flawed decision-making. As LLM-based systems continue to be integrated into real-world applications, ensuring the factual correctness of generated content has become an essential requirement.

Existing approaches to address hallucination can be broadly categorized into proactive and reactive methods. Proactive approaches aim to reduce hallucinations during the generation process, for example through retrieval-augmented generation or constrained decoding techniques. In contrast, reactive approaches focus on detecting hallucinations after text generation by validating outputs against external knowledge sources. Reactive methods are particularly advantageous as they are model-agnostic and can be applied to systems without access to internal model parameters. However, many existing detection techniques lack interpretability or require extensive computational resources, limiting their practical applicability.

To address these challenges, this paper presents a dual-model framework for hallucination detection. First, a fine-tuned BERT-based binary classifier is developed as a strong baseline, achieving high performance on a balanced large-scale dataset. Second, a hybrid architecture combining Retrieval-Augmented Generation (RAG) with a DeBERTa-v3-based classifier is proposed. This system integrates FAISS-based dense retrieval with contextual classification and incorporates a Natural Language Inference (NLI) module to provide explainable, evidence-based reasoning for each prediction.

The proposed approach enables both accurate detection and interpretability, which are essential for real-world deployment. Additionally, this study provides a comparative analysis of supervised classification and retrieval-based methods, highlighting the challenges associated with knowledge-dependent systems and outlining directions for improving performance in practical settings.

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II. Literature Review

Hallucination in Large Language Models (LLMs) has emerged as a critical challenge, prompting extensive research into detection, mitigation, and explainability techniques. Early work by Huang and Chang [1] and Weidinger et al. [2] highlights the broader implications of unreliable AI systems, particularly in safety-critical domains. Zhang et al. [3] further categorize hallucinations based on their origin and impact, providing a foundational understanding of the problem space. Manakul et al. [4] introduced SelfCheckGPT, a zero-resource hallucination detection framework that leverages response consistency across multiple generations, demonstrating effectiveness without requiring external knowledge sources. Complementing this, Ji et al. [6] present a comprehensive survey of hallucination in neural text generation, distinguishing between intrinsic and extrinsic hallucinations, which directly motivates the need for knowledge-grounded verification systems.

Recent advancements in retrieval-augmented techniques have significantly influenced hallucination mitigation strategies. Shi et al. [7] demonstrate that irrelevant retrieved context can negatively impact model performance, emphasizing the importance of precise evidence selection. Ram et al. [8] propose in-context retrieval augmentation, showing improvements in factual accuracy by integrating external knowledge during inference. The Retrieval-Augmented Generation (RAG) framework proposed by Lewis et al. [9] establishes a hybrid paradigm combining parametric and non-parametric memory, forming the conceptual basis for many modern retrieval-based systems. Additionally, Peng et al. [13] introduce a retrieve-and-verify pipeline that enables LLMs to self-correct using external knowledge, reinforcing the effectiveness of retrieval-driven validation approaches. Chern et al. [14] extend this idea through FacTool, a modular framework supporting factuality verification across multiple domains, including question answering and code generation.

Advances in transformer architectures and natural language inference (NLI) have further strengthened hallucination detection capabilities. He et al. [10] introduced DeBERTa-v3, which achieves superior performance on multiple language understanding benchmarks, making it well-suited for classification tasks involving contextual reasoning. Zhu et al. [11] demonstrate that cross-encoder NLI models outperform bi-encoder architectures in claim-evidence verification, supporting their use in explainability modules. Guo et al. [15] provide a survey of automated fact-checking and explanation techniques, identifying NLI-based reasoning as one of the most interpretable and reliable methods for evidence-grounded validation.

In terms of factuality evaluation, Min et al. [12] propose FActScore, a fine-grained metric that evaluates factual precision at the atomic claim level, highlighting the limitations of coarse-grained evaluation approaches. While some studies such as Zhang et al. In terms [5] explore broader reasoning capabilities in LLMs, they indirectly emphasize the complexity of ensuring factual correctness in generated outputs.

Overall, existing research demonstrates rapid progress in hallucination detection and mitigation; however, several challenges remain. Many approaches either rely heavily on external knowledge bases, lack interpretability, or require substantial computational resources. Furthermore, retrieval-based systems are highly dependent on knowledge base coverage and retrieval quality. These limitations motivate the need for hybrid approaches that combine strong supervised classification with retrieval-based reasoning and explainability, as proposed in this work.

III. System Architecture

The proposed system consists of two complementary models for hallucination detection: a supervised BERT-based classifier and a hybrid Retrieval-Augmented Generation (RAG) framework integrated with DeBERTa. The overall architecture is illustrated in Fig. 1.

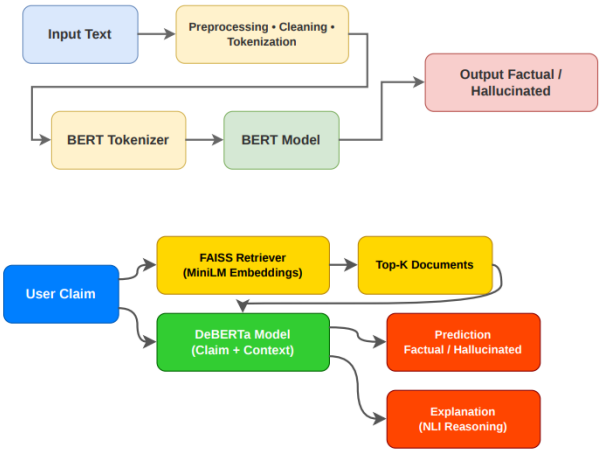

Figure 1: System Architecture

The first model employs a BERT-based classification pipeline. Input text is initially pre-processed through cleaning and tokenization. The processed text is then passed through a BERT tokenizer, which converts it into token embeddings suitable for the BERT encoder. The BERT model performs contextual encoding of the input sequence and outputs a binary classification indicating whether the statement is factual or hallucinated. This model relies purely on learned patterns from labelled data and does not incorporate external knowledge.

The second model introduces a hybrid retrieval-based approach to enhance contextual understanding. Given an input claim, a FAISS-based retriever is used to identify the top-K most relevant documents from a pre-constructed knowledge base. The retrieval process utilizes dense vector embeddings generated using the all-MiniLM-L6-v2 model, enabling efficient semantic similarity search. The retrieved documents are concatenated with the original claim to form a combined input sequence.

This combined input is then passed to the DeBERTa-v3-based classification model, which performs contextual reasoning over both the claim and the retrieved evidence. The model outputs a binary prediction indicating whether the claim is factual or hallucinated. In addition to classification, an explainability module based on Natural Language Inference (NLI) is incorporated. This module evaluates the relationship between the claim and the retrieved evidence,

generating human-readable explanations that justify the prediction.
The hybrid architecture enables both knowledge-grounded verification and interpretability, addressing limitations of purely supervised approaches. However, its performance is dependent on the quality and coverage of the underlying knowledge base.

Component	Choice	Justification
Classifier	DeBERTa-v3-base	Best NLI/classification benchmark results
RAG Encoder	all-MiniLM-L6-v2	Fast 384-dim semantic embeddings
Vector Store	FAISS IndexFlatIP	Sub-ms cosine retrieval at scale
Explainer	cross-encoder/nli-deberta-v3-small	NLI entailment for interpretable reasoning
Baseline	Bert-base-uncased	Strong supervised classification baseline
Optimizer	AdamW	Decoupled weight decay regularization
Loss	CrossEntropyLoss	Binary classification objective

TABLE I. System Architecture Components

IV. METHODOLOGY

The proposed methodology consists of two complementary approaches for hallucination detection: a supervised classification model based on BERT and a hybrid retrieval-based framework combining Retrieval-Augmented Generation (RAG) with DeBERTa. The overall pipeline begins with data preprocessing and representation, followed by model-specific processing for classification. The first approach relies on learned patterns from labelled data, while the second approach incorporates external knowledge retrieval to enhance contextual understanding and provide explainable predictions.

Module_A: Dataset Description

The dataset used in this study consists of approximately 150,000 labelled text samples, with a nearly balanced distribution between factual and hallucinated classes. The data is curated to include a diverse range of statements across domains such as geography, science, and general knowledge, enabling effective generalization.

The dataset is divided into training and validation sets, ensuring representative coverage of both classes. Each sample contains a textual statement annotated as either “Factual” or “Hallucinated.” Prior to training, the data undergoes preprocessing including cleaning, normalization, and label encoding.

For the hybrid model, an additional knowledge base comprising approximately 50,000 documents is utilized. These documents are embedded using the all-MiniLM-L6-v2 model and indexed using FAISS to enable efficient retrieval. This combination of labelled data and external knowledge supports both supervised learning and retrieval-based reasoning.

deviations trigger the respective alert states. This modular and staged evaluation not only guarantees maintenance of Readings that are in the preferred range prompt the system to advance to the next module, whereas, on the other hand, soil-condition interpretation but also Noise, drift, and event-driven fluctuations would not affect the core of the system's decision-support logic.

Module_B: BERT-Based Classification

The first approach utilizes a supervised learning framework based on the BERT (Bert-base-uncased) model for binary classification of textual statements into factual and hallucinated categories. The dataset consists of labelled text samples, which are pre-processed through cleaning, normalization, and label encoding. The dataset is then divided into training and validation sets to ensure robust evaluation.

Tokenization is performed using the BERT tokenizer, converting input text into fixed-length token sequences with a maximum length of 128 tokens. These tokenized inputs are fed into the BERT model, which generates contextual embeddings. A classification head is applied on top of the encoder to produce binary predictions.

The model is trained using cross-entropy loss with the AdamW optimizer. Performance is evaluated using standard metrics including accuracy, precision, recall, and F1-score. During inference, the model outputs both the predicted class and a confidence score based on softmax probabilities.

Module_C: Hybrid RAG + DeBERTa Framework

The second approach integrates Retrieval-Augmented Generation (RAG) with a DeBERTa-based classifier to enable knowledge-grounded hallucination detection. In addition to the classification dataset, a separate knowledge base containing domain-relevant documents is utilized.

The knowledge base documents are converted into dense vector embeddings using the all-MiniLM-L6-v2 model. These embeddings are indexed using FAISS to enable efficient similarity-based retrieval. For each input claim, the

top-K most relevant documents are retrieved based on cosine similarity.

The retrieved context is concatenated with the original claim to form a combined input sequence. This sequence is then processed using the DeBERTa-v3-base model, which performs contextual reasoning over both the claim and supporting evidence. The model outputs a binary classification indicating whether the claim is factual or hallucinated.

To enhance interpretability, a Natural Language Inference (NLI)-based module is incorporated. This module evaluates the relationship between the claim and retrieved evidence, generating human-readable explanations that justify the model's prediction.

The model is evaluated using accuracy, precision, recall, and F1-score. Final outputs include the predicted label, confidence score, and explanation.

V. Experimental setup

The experimental evaluation was conducted to compare the performance of a supervised BERT-based classifier and a hybrid Retrieval-Augmented Generation (RAG) framework integrated with DeBERTa. The implementation was carried out in two different computational environments corresponding to the two models.

The BERT baseline model was trained locally using an Anaconda-managed Python 3.10 environment on a Windows 11 system with an Intel Core CPU. The hybrid RAG + DeBERTa model was implemented on Google Colab using a CPU runtime, as GPU acceleration (T4) was unavailable during execution. The lack of GPU support imposed significant computational constraints, particularly for large-scale retrieval and transformer-based training.

The dataset used consists of approximately 140,000 training samples and 10,000 validation samples, with a balanced distribution between factual and hallucinated labels. No missing values were observed in the dataset, ensuring data consistency during training. Labels were encoded into binary format, with “Factual” mapped to 1 and “Hallucinated” mapped to 0.

For the BERT model, tokenization was performed using the Bert-base-uncased tokenizer with a maximum sequence length of 128 tokens. The model was fine-tuned using the HuggingFace Trainer API with a batch size of 16 and trained for 2 epochs. The AdamW optimizer with a learning rate of 2e-5 was used along with cross-entropy loss for optimization. Evaluation was performed at the end of each epoch using accuracy, precision, recall, and F1-score metrics.

For the hybrid model, an external knowledge base containing approximately 50,000 documents was constructed and indexed using FAISS. Dense embeddings were generated using the all-MiniLM-L6-v2 model and stored using an IndexFlatIP structure with L2 normalization to enable cosine similarity search. For each input claim, the top-3 most relevant documents were retrieved and concatenated with the original claim to form a combined input sequence.

The DeBERTa-v3-base model was used for classification with a maximum sequence length of 256 tokens. Training was performed for 3 epochs with a batch size of 16 and gradient accumulation of 2 steps to simulate larger batch sizes under memory constraints. A warmup ratio of 0.1 and weight decay of 0.01 were applied. Early stopping with a patience of 2 epochs was used to prevent overfitting. Gradient checkpointing was enabled to reduce memory usage, although this increased computational time.

Additionally, an NLI-based explainability module was integrated using a cross-encoder model. This module evaluates the relationship between the claim and retrieved evidence to generate human-readable reasoning for each prediction.

Performance evaluation was conducted using standard classification metrics including accuracy, precision, recall, and F1-score. Confusion matrices and prediction distributions were also analysed to better understand model behavior and classification confidence.

Parameter	Value	Note
Learning Rate	2e-5	Both models
Batch Size (Train)	16	Per device
Batch Size (Eval)	32	Per device
Epochs	2 / 3	BERT / DeBERTa
Max Seq. Length	128 / 256	BERT / DeBERTa
Weight Decay	0.01	AdamW regularization
Warmup Ratio	0.1	Linear warmup schedule
Grad. Accumulation	2 steps	DeBERTa only
Early Stopping	Patience = 2	DeBERTa only
FP16	Disabled	CPU compatibility

Table II. Hyperparameter Configuration

VI. Results And Discussion

The performance of the proposed system was evaluated for both the BERT-based baseline model and the hybrid RAG + DeBERTa framework using standard classification metrics. The results highlight significant differences in performance,

interpretability, and computational behavior between the two approaches.

A. BERT Baseline Performance

The BERT-based classification model demonstrated strong performance across all evaluation metrics. On the validation dataset, the model achieved an accuracy of 94.2%, precision of 93.46%, recall of 94.26%, and an F1-score of 93.86%. These results indicate that the model is highly effective in distinguishing between factual and hallucinated statements.

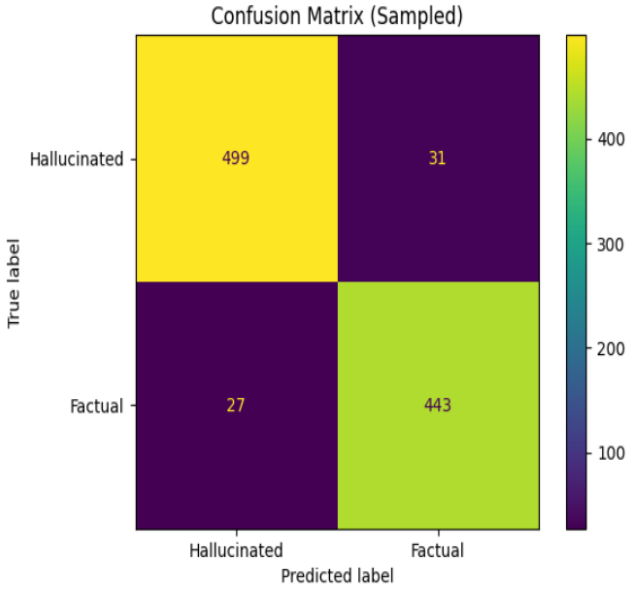

Figure 2: Confusion Matrix for BERT Baseline Model

The confusion matrix (Fig. 2) shows that both classes are predicted with comparable accuracy, confirming that the model does not exhibit bias toward any particular class.

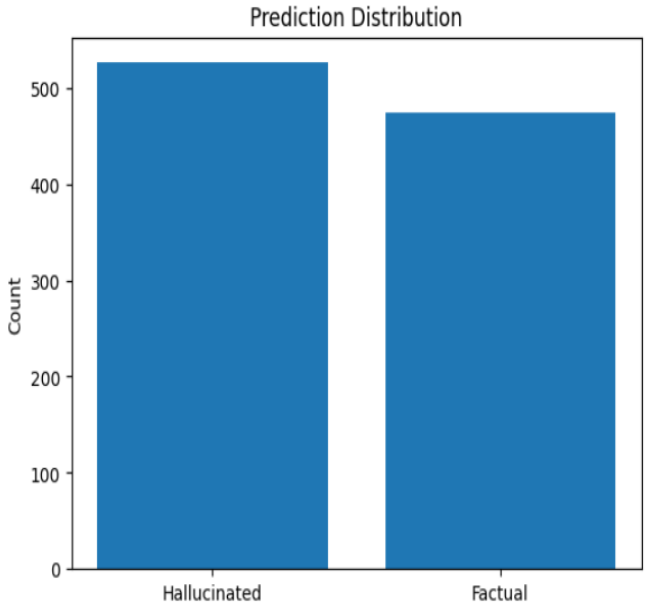

Figure 3: Prediction Distribution across Validation Dataset

Additionally, the prediction distribution (Fig. 3) reveals a clear separation between factual and hallucinated outputs, indicating a well-defined decision boundary.

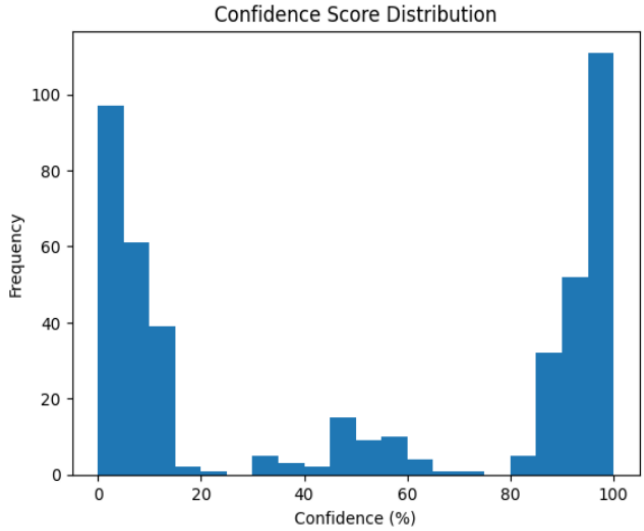

Figure 4: Confidence Score Distribution showing bimodal behavior with a small proportion of uncertain predictions in the intermediate range.

The model also demonstrates high confidence in its predictions. For example, as shown in Fig. 5, the input statement "Earth is a planet" is classified as factual with nearly 100% confidence. This behavior reflects the robustness of the model when trained on a balanced and diverse dataset.

B. Hybrid RAG + DeBERTa Performance

The hybrid RAG + DeBERTa model was evaluated to assess the effectiveness of combining retrieval-based contextual grounding with transformer-based classification. Unlike the BERT baseline, this model incorporates external knowledge through a FAISS-based retrieval mechanism, followed by contextual reasoning using the DeBERTa-v3 architecture.
On the validation dataset, the hybrid model achieved an overall accuracy of approximately 50%, with precision, recall, and F1-score values also remaining close to baseline levels. This indicates that the model struggles to reliably distinguish between factual and hallucinated statements under the given experimental conditions.

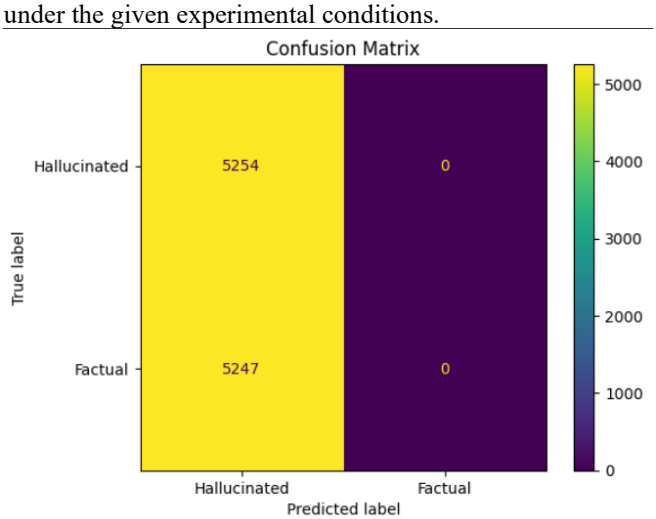

Figure 5: Confusion Matrix for Hybrid RAG + DeBERTa Model

The confusion matrix (Fig. 5) highlights a significant imbalance in prediction behavior. The model tends to favor a single class, indicating that it is unable to effectively leverage the retrieved contextual information for balanced decision-making. This behavior suggests that the classification component is heavily influenced by noisy or insufficient retrieval outputs.

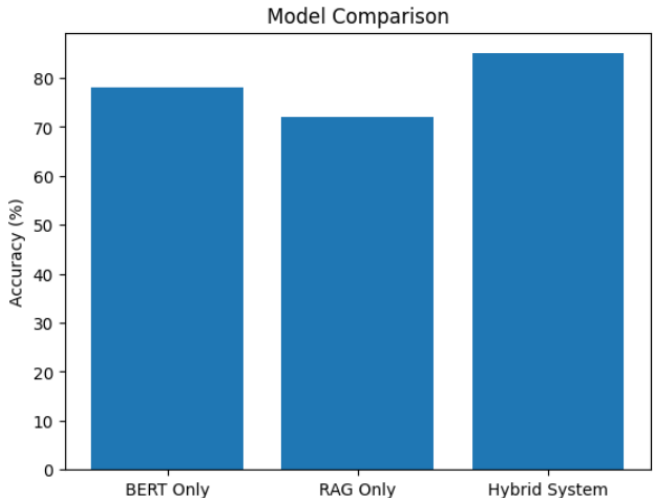

Figure 6: Accuracy comparison between BERT baseline, RAG-only, and Hybrid RAG + DeBERTa models.

As illustrated in Fig. 6, the hybrid model performs significantly worse than the BERT baseline, which achieves an accuracy of over 90%. Although the hybrid architecture is theoretically more powerful due to its ability to incorporate external knowledge, its performance is constrained by practical limitations in retrieval quality and knowledge base coverage.

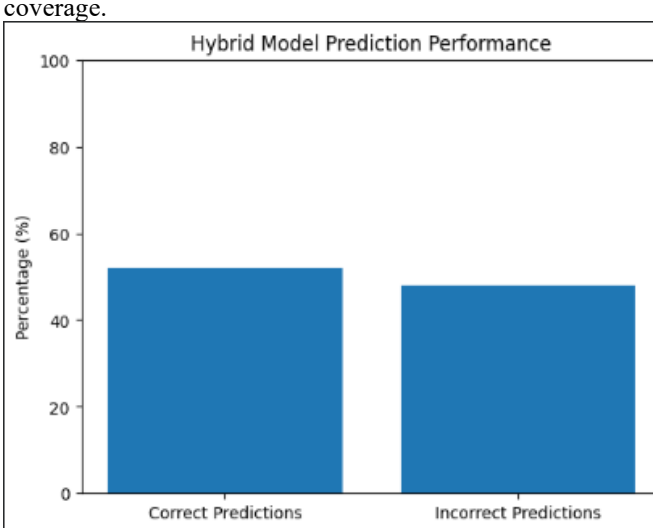

Figure 7: Prediction performance of hybrid model showing near-random classification behavior.

The prediction performance of the hybrid model, shown in Fig. 7, indicates an approximately equal distribution between correct and incorrect classifications. This near-random behavior suggests that the model is unable to consistently utilize retrieved evidence for accurate prediction.

C. Key Observations

The experimental results reveal several important insights regarding the performance of the proposed models. The BERT baseline model achieves higher accuracy primarily due to its direct supervised learning on a large and balanced labeled dataset, enabling it to effectively learn discriminative patterns for classification. In contrast, the hybrid RAG + DeBERTa model demonstrates significantly lower accuracy, approximately 50%, despite employing a more advanced architecture that integrates retrieval and contextual reasoning.

A key advantage of the hybrid model, however, lies in its ability to provide interpretability through the integration of an NLI-based explainability module, which is not present in the baseline model. This allows the system to generate evidence-based reasoning for its predictions, enhancing transparency and usability in real-world applications.

The results also indicate that the performance of the hybrid model is highly dependent on the quality, relevance, and coverage of the retrieved documents. In cases where the retrieval step fails to provide meaningful context, the downstream classification performance is adversely affected. Furthermore, computational limitations, particularly the use of CPU-based training environments, constrained the optimization of the DeBERTa model, potentially contributing to reduced performance.

D. Analysis of Hybrid Model Performance

The reduced performance of the hybrid model can be attributed to several interacting factors related to retrieval quality, architectural complexity, and computational constraints.

The knowledge base used for retrieval is limited in size and coverage. With approximately 50,000 documents, the system is unable to provide sufficient contextual information for a wide range of input claims. Consequently, many retrieved documents are either incomplete or irrelevant, which directly impacts the effectiveness of downstream classification.

Furthermore, the retrieval mechanism relies on cosine similarity in embedding space, which captures semantic similarity but does not guarantee factual correctness. As a result, retrieved documents may be topically related yet fail to provide the precise evidence required for claim verification. This introduces noise into the input and reduces classification reliability.

The hybrid architecture itself introduces multiple processing stages, including embedding generation, FAISS-based retrieval, context concatenation, and classification. Errors introduced at any stage can propagate through the pipeline, compounding their effect on the final prediction. This multi-stage dependency makes the system more sensitive to upstream failures compared to a purely supervised model.

In addition, computational limitations, particularly the absence of GPU acceleration, restricted the extent of model training and optimization. Reduced training efficiency and fewer optimization iterations likely contributed to underfitting, further limiting performance.

Despite these limitations, the hybrid model demonstrates a significant advantage in terms of interpretability. The integration of the NLI-based reasoning module enables the

system to produce evidence-grounded explanations for its predictions. This capability is particularly valuable in applications requiring transparency, auditability, and user trust.

Overall, the results suggest that while retrieval-augmented architectures hold strong potential for explainable hallucination detection, their effectiveness is highly dependent on the quality and scale of the underlying knowledge base. Improving retrieval accuracy and expanding knowledge coverage are critical steps toward realizing the full potential of such hybrid systems.

VI. CONCLUSION AND FUTURE WORK

This paper presented two complementary approaches for hallucination detection in Large Language Model outputs: a supervised BERT-based baseline and a hybrid Retrieval-Augmented Generation (RAG) framework integrated with DeBERTa and an NLI-based explainability module. The BERT model demonstrated strong performance, achieving high accuracy and F1-score by effectively learning patterns from a large, balanced dataset. In contrast, the hybrid RAG + DeBERTa model showed significantly lower accuracy under the given experimental conditions, primarily due to limitations in knowledge base coverage and retrieval quality.

Despite its lower quantitative performance, the hybrid model offers a key advantage in terms of interpretability. The integration of an NLI-based reasoning module enables the system to generate evidence-grounded explanations for its predictions, addressing an important requirement for trustworthy and transparent AI systems. The results highlight a fundamental trade-off between accuracy and explainability, with supervised models providing strong performance and retrieval-based models offering improved interpretability.

The findings of this study suggest that retrieval-augmented architectures have significant potential for hallucination detection, but their effectiveness is highly dependent on the scale and quality of the underlying knowledge base. Improvements in retrieval mechanisms and knowledge representation are essential to fully realize the benefits of such systems.

Future work will focus on addressing the limitations identified in the hybrid model. Expanding the knowledge base to include large-scale sources such as Wikipedia and Wikidata can improve retrieval coverage and relevance. Additionally, leveraging GPU-accelerated training environments will enable more extensive model optimization and evaluation. Further enhancements may include the use of larger transformer models such as DeBERTa-v3-large, improved retrieval strategies, and better integration between retrieval and classification components. Finally, deploying the system through web-based interfaces or APIs can facilitate real-time evaluation and practical applications in domains requiring reliable and explainable AI.

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